Audist

ASHRAE LEVEL 2 ENERGY AUDIT

SYNTHETIC TEST - CVS-Style Pharmacy

Test City, California

ASHRAE Level 2 Energy Audit Report

Prepared for	Software acceptance testing only
Prepared by	Audist sample report workflow
Audit date	2026-08-20
Report date	2026-08-24
Project number	SYNTHETIC-V7
Revision	0

SYNTHETIC TEST REPORT - NOT FOR CLIENT USE

Document control Report integrity: PASS_WITH_WARNINGS

REPORT NAVIGATION

Table of Contents

Technical Appendices

01 Executive Summary

This synthetic report demonstrates Audist's professional reporting system. All utility, equipment, ECM, cost, and savings values are fabricated for software and visual acceptance testing; they are not verified engineering results.

FACILITY SIZE **25,000 ft^2**	ANNUAL ELECTRICITY **Not Calculated**	ANNUAL NATURAL GAS **Not Calculated**
ANNUAL UTILITY COST **Not Calculated**	SITE EUI **Not Calculated**	PEAK DEMAND **71 kW**
RECOMMENDED ECMS **1**	RECOMMENDED PORTFOLIO ELECTRIC SAVINGS **25,000 kWh/yr**	RECOMMENDED PORTFOLIO ANNUAL COST SAVINGS **$2,500**
RECOMMENDED PORTFOLIO IMPLEMENTATION COST **$5,000**	RECOMMENDED PORTFOLIO SIMPLE PAYBACK **2 yr**	

ECM Executive Summary

ECM	Measure	Electric Savings	Annual Cost Savings	Project Cost	Payback	Decision
ECM-01	HVAC Schedule Optimization	25,000 kWh/yr	$2,500	$5,000	2 yr	Recommended
ECM-02	Economizer Repair	Not Calculated	Not Calculated	Not Calculated	Not Calculated	Further Investigation
ECM-03	Heat Pump Water Heater	1,000 kWh/yr	$1,000	$20,000	20 yr	Not Recommended

02 Facility Description

The facility description is limited to documented site and equipment records. Attributes not present in the audit are not inferred.

Facility Summary

Attribute	Value
Facility Type	Retail / Pharmacy
Floor Area	25,000 ft^2
Year Built	1998
Operating Schedule	100 hr/week
Primary Utility Providers	Test Utility
Audit Date	2026-08-20

03 Audit Scope and Methodology

Systems documented as present: Packaged HVAC, Refrigeration. Evidence classifications represented include Measured. Audist distinguishes Measured, Nameplate, Manufacturer, BAS / Trend, Utility Bill, Calculated, Estimated, and Assumed evidence. Calculations retain method/version, inputs, units, provenance, assumptions, and QA; deterministic QA remains separate from optional AI review.

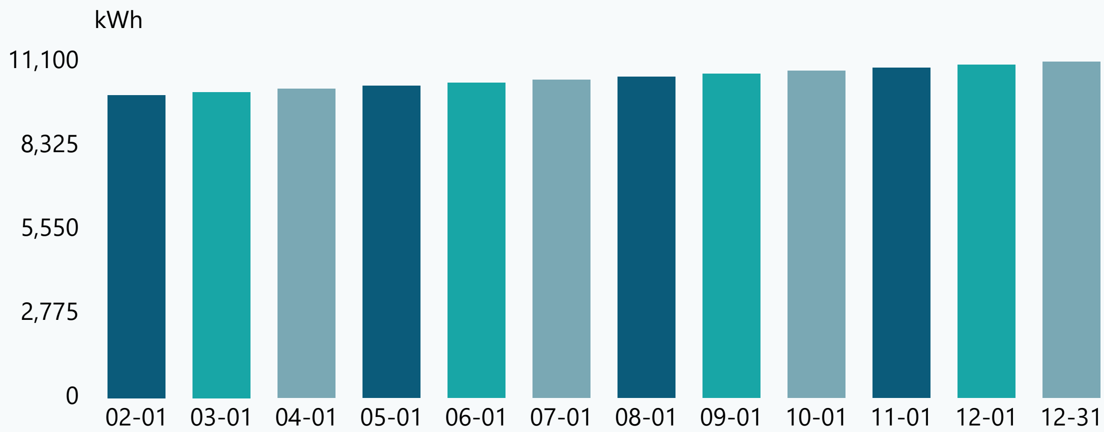
04 Utility Analysis

Utility Summary

Utility	Coverage	Annual Usage	Peak Demand	Annual Cost
Electricity	Partial	Not Calculated	71 kW	Not Calculated

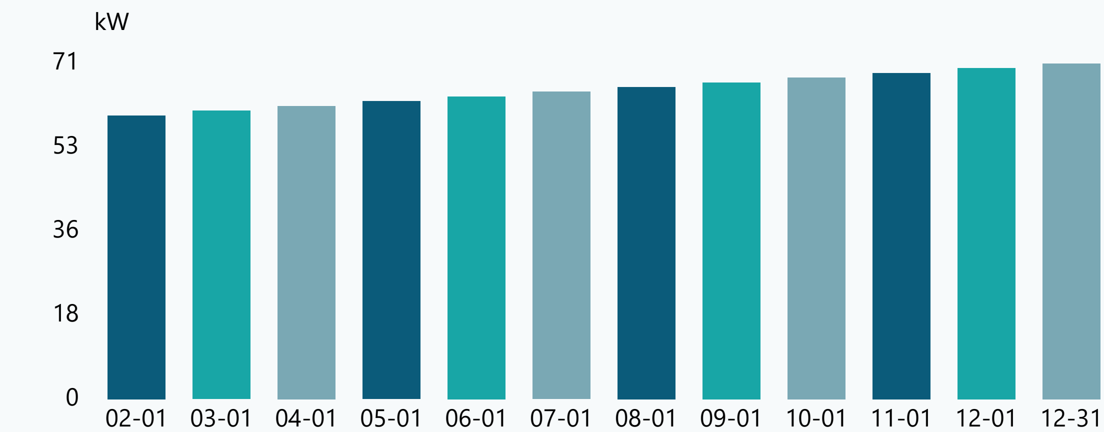
Monthly Electricity Consumption

Annual total across displayed periods: 126,600 kWh



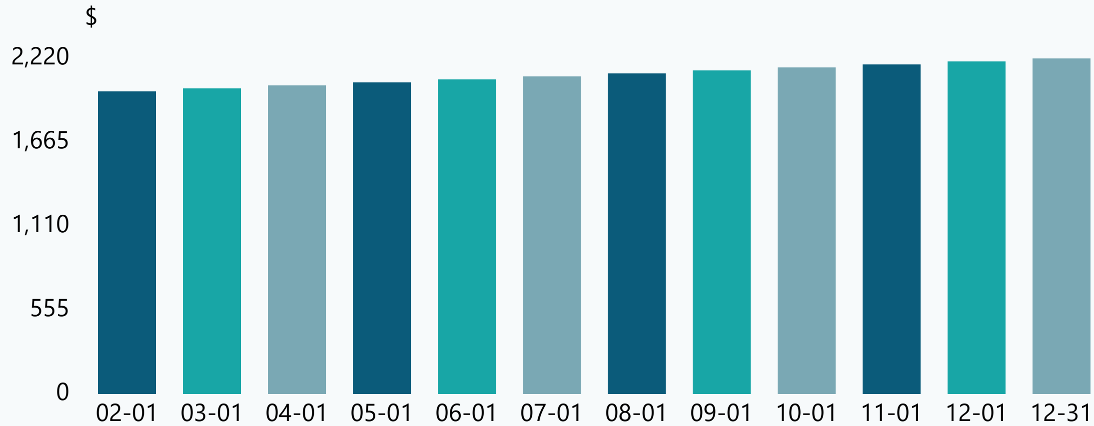
Monthly Peak Demand

Annual total across displayed periods: 786 kW



Monthly Utility Cost

Annual total across displayed periods: $25,320



05 Existing Building Systems

Packaged HVAC

ID	Type	Manufacturer / Model	Capacity / Rating	Controls	Schedule	Condition
RTU-01	RTU	Acme / 10T	10 tons	Economizer	100 hr/week	—

Refrigeration

ID	Type	Manufacturer / Model	Capacity / Rating	Controls	Schedule	Condition
REF-01	Pharmacy / Medical Refrigerator	ColdCo / RX	—	—	—	—

06 Energy-Use / End-Use Analysis

Modeled End Uses

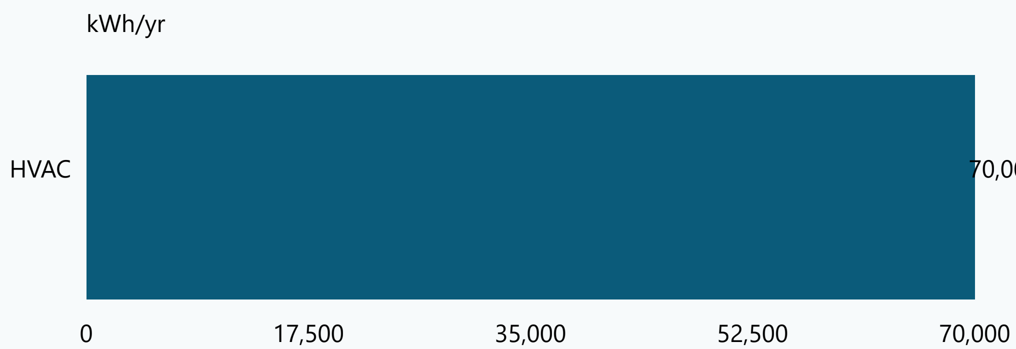
Utility	End Use	Annual Energy	% Facility	Evidence
Electricity	HVAC	70,000 kWh/yr	Not Calculated	C

Utility-to-Model Reconciliation

Utility	Utility Baseline	Modeled Energy	Unmodeled / Unexplained	Gap	Status
Electricity	Not Calculated	70,000 kWh/yr	Not Calculated	Not Calculated	NO_BASELINE

Modeled End-Use Breakdown

Ranked modeled consumption; reconciliation gap remains explicit in the table below.



07 Energy Conservation Measures

ECM-01

RECOMMENDED

HVAC Schedule Optimization

25,000 kWh/yr	$2,500	$5,000	2 yr
ELECTRIC SAVINGS	COST SAVINGS / YR	PROJECT COST	SIMPLE PAYBACK

EVIDENCE QUALITY
B

CALCULATION STAGE
ENGINEERING ESTIMATE

AFFECTED EQUIPMENT
RTU-01

Existing Condition

Extended operation was documented relative to the recorded facility schedule.

Recommended Improvement

Reduce HVAC operation after confirming occupied ventilation and control requirements.

Energy Analysis

CALCULATION METHOD **CALC-HVAC-001 v1.0**

ANNUAL SAVINGS
Operating kW x avoided annual hours

Calculation Inputs — c1

Input	Value	Unit	Provenance
Operating power	10	kW	Measured

Assumptions & Limitations

Schedule must be verified before implementation

Implementation Considerations

Maintain ventilation and humidity requirements

Measurement & Verification

Verify installed condition and relevant operating parameters; M&V has not been completed.

ECM-02

FURTHER INVESTIGATION

Economizer Repair

Not Calculated	Not Calculated	Not Calculated	Not Calculated
ELECTRIC SAVINGS	COST SAVINGS / YR	PROJECT COST	SIMPLE PAYBACK

EVIDENCE QUALITY
NOT ASSESSED

CALCULATION STAGE
NOT ASSESSED

AFFECTED EQUIPMENT
RTU-01

Existing Condition

A suspected economizer malfunction requires functional testing.

Recommended Improvement

Investigate, repair, and commission the economizer sequence.

Energy Analysis

Further Investigation

Annual savings are not quantified. Additional engineering inputs or a validated method are required.

Implementation Considerations

To be developed during design.

Measurement & Verification

Verify installed condition and relevant operating parameters; M&V has not been completed.

ECM-03

NOT RECOMMENDED

Heat Pump Water Heater

1,000 kWh/yr	$1,000	$20,000	20 yr
ELECTRIC SAVINGS	COST SAVINGS / YR	PROJECT COST	SIMPLE PAYBACK

EVIDENCE QUALITY
NOT ASSESSED

CALCULATION STAGE
NOT ASSESSED

AFFECTED EQUIPMENT
Not linked

Existing Condition

Synthetic comparison case.

Recommended Improvement

Replace the existing water heater with a heat-pump water heater.

Energy Analysis

CALCULATION METHOD **CALC-DHW-002 v1.0**

ANNUAL SAVINGS
See detailed calculation appendix.

Implementation Considerations

To be developed during design.

Measurement & Verification

Verify installed condition and relevant operating parameters; M&V has not been completed.

08 ECM Portfolio Summary

ECM Portfolio Summary

Portfolio	Status	Electric Savings	Fuel Savings	Demand Savings	Annual Cost Savings	Implementation Cost	Simple Payback	Evidence / Maturity
Recommended Portfolio	CALCULATED	25,000 kWh/yr	0 therms/yr	0 kW	$2,500	$5,000	2 yr	A / ENGINEERING ESTIMATE

09 Financial Analysis

ECM Financial Summary

ECM	Recommendation	Annual Cost Savings	Implementation Cost	Simple Payback
ECM-01	Recommended	$2,500	$5,000	2 yr
ECM-02	Further Investigation	Not Calculated	Not Calculated	Not Calculated
ECM-03	Not Recommended	$1,000	$20,000	20 yr

10 Implementation Considerations

Implementation requirements are planning considerations, not final design documents. Verify scope, controls, code, safety, operations, and procurement before construction.

11 Measurement & Verification Considerations

M&V considerations are recommendations only. Audist does not imply that post-installation verification has occurred.

12 Assumptions and Limitations

Limitation

No weather normalization

Limitation

No source-energy EUI

Limitation

No end-use disaggregation

Limitation

Partial years are not annualized

Limitation

Annual reconciliation only

Limitation

No automatic residual allocation

Limitation

No automatic scaling

Limitation

No ECM portfolio interaction adjustment

Limitation

Standalone ECM savings are not assumed additive

Limitation

Only engineer-confirmed sequential remaining-baseline adjustments are calculated

Limitation

No generic interaction percentages

Limitation

No automatic portfolio optimization

Limitation

No automatic lighting/HVAC interaction

Limitation

ECM DEMAND EXCEEDS OBSERVED PEAK

Limitation

Calculation input is not traceable

Limitation

Calculated ECM has no affected equipment

Limitation

Simple payback lacks valid cost and savings

Limitation

Avoided hours documented

Limitation

ECM-02 — Economizer Repair: savings are not quantified.

13 Conclusions / Recommended Next Steps

> Recommended next steps are to resolve material limitations, complete engineer review, validate proposed conditions and costs, and issue only measures explicitly approved in this report.

TECHNICAL APPENDIX

Appendix A — Detailed Equipment Inventory

Detailed Equipment Inventory

ID	System	Type	Manufacturer / Model	Capacity / Rating	Notes
RTU-01	PackagedHVAC	RTU	Acme / 10T	10 tons	—
REF-01	Refrigeration	Pharmacy / Medical Refrigerator	ColdCo / RX	—	—

TECHNICAL APPENDIX

Appendix B — Measurements

Measurements

Equipment	Parameter	Value	Unit	Provenance	Method / Instrument	Timestamp
RTU-01	Supply fan power	2.2	kW	Measured	Power meter	2026-08-20T10:00:00Z

TECHNICAL APPENDIX

Appendix C — Utility Data

Utility Data

Utility	Provider / Account	Billing Period	Usage	Unit	Peak kW	Cost	Source
Electricity	Test Utility / Main	2025-01-01 to 2025-02-01	10000	kWh	60	2000	Utility Bill
Electricity	Test Utility / Main	2025-02-01 to 2025-03-01	10100	kWh	61	2020	Utility Bill
Electricity	Test Utility / Main	2025-03-01 to 2025-04-01	10200	kWh	62	2040	Utility Bill
Electricity	Test Utility / Main	2025-04-01 to 2025-05-01	10300	kWh	63	2060	Utility Bill
Electricity	Test Utility / Main	2025-05-01 to 2025-06-01	10400	kWh	64	2080	Utility Bill
Electricity	Test Utility / Main	2025-06-01 to 2025-07-01	10500	kWh	65	2100	Utility Bill
Electricity	Test Utility / Main	2025-07-01 to 2025-08-01	10600	kWh	66	2120	Utility Bill
Electricity	Test Utility / Main	2025-08-01 to 2025-09-01	10700	kWh	67	2140	Utility Bill
Electricity	Test Utility / Main	2025-09-01 to 2025-10-01	10800	kWh	68	2160	Utility Bill
Electricity	Test Utility / Main	2025-10-01 to 2025-11-01	10900	kWh	69	2180	Utility Bill
Electricity	Test Utility / Main	2025-11-01 to 2025-12-01	11000	kWh	70	2200	Utility Bill
Electricity	Test Utility / Main	2025-12-01 to 2025-12-31	11100	kWh	71	2220	Utility Bill

TECHNICAL APPENDIX

Appendix D — Detailed Calculations

Detailed Calculations

Calculation	ECM	Method / Version	Status	Inputs / Provenance	Outputs
c1	ECM-01	CALC-HVAC-001 v1.0	Calculated	Operating power: 10 kW [Measured]	annualKwhSavings: 25000 kWh/yr; annualCostSavings: 2500 $/yr
c3	ECM-03	CALC-DHW-002 v1.0	Calculated	—	annualKwhSavings: 1000 kWh/yr; annualCostSavings: 1000 $/yr

TECHNICAL APPENDIX

Appendix E — Photo Log

Photo Log

Photo ID	Equipment	Category	Caption / Note	Captured At

TECHNICAL APPENDIX

Appendix F — QA/QC Findings

QA/QC Findings

Finding	Severity	Status	Category / Finding	Description / Engineer Note
QA-UTIL-001-11hyi7m	MEDIUM	OPEN	UTILITY_QA - Utility baseline is incomplete	One or more utility accounts do not form a complete annual baseline.
QA-UTIL-004-19ompgs	HIGH	OPEN	UTILITY_QA - ECM DEMAND EXCEEDS OBSERVED PEAK	Claimed savings exceed the applicable observed facility boundary.
QA-UTIL-004-88oo86	HIGH	OPEN	UTILITY_QA - ECM DEMAND EXCEEDS OBSERVED PEAK	Claimed savings exceed the applicable observed facility boundary.
QA-CALC-003-1swwvdm	HIGH	OPEN	PROVENANCE_QA - Calculation input is not traceable	c1 input Operating power lacks provenance or source description.
QA-ENDUSE-002-kbtz44	MEDIUM	OPEN	END_USE_RECONCILIATION - MAJOR SYSTEM NOT MODELED	MAJOR_SYSTEM_NOT_MODELED
QA-ECM-003-5celu7	HIGH	OPEN	ECM_QA - Calculated ECM has no affected equipment	ECM-03 has a calculation or savings claim without an equipment relationship.
QA-ECM-004-k60xnn	MEDIUM	OPEN	ECM_QA - Implementation cost lacks provenance	ECM-01 has an implementation cost without a source or basis.
QA-ECM-004-1cyjrs5	MEDIUM	OPEN	ECM_QA - Implementation cost lacks provenance	ECM-03 has an implementation cost without a source or basis.
QA-ECM-005-wos5tb	MEDIUM	OPEN	ECM_QA - Calculated ECM has no visible assumptions	ECM-03 uses a calculated result with no assumption record.
QA-ECON-002-18spz23	HIGH	OPEN	ECONOMICS_QA - Simple payback lacks valid cost and savings	ECM-01 reports payback without positive annual cost savings and a valid implementation cost.
QA-ECON-002-wbfrhl	HIGH	OPEN	ECONOMICS_QA - Simple payback lacks valid cost and savings	ECM-03 reports payback without positive annual cost savings and a valid implementation cost.

TECHNICAL APPENDIX

Appendix G — Assumptions / Supporting Data

Assumptions / Supporting Data

Source	Assumption / Limitation
L-1	No weather normalization
L-2	No source-energy EUI
L-3	No end-use disaggregation
L-4	Partial years are not annualized
L-5	Annual reconciliation only
L-6	No automatic residual allocation
L-7	No automatic scaling
L-8	No ECM portfolio interaction adjustment
L-9	Standalone ECM savings are not assumed additive
L-10	Only engineer-confirmed sequential remaining-baseline adjustments are calculated
L-11	No generic interaction percentages
L-12	No automatic portfolio optimization
L-13	No automatic lighting/HVAC interaction
L-14	ECM DEMAND EXCEEDS OBSERVED PEAK
L-15	Calculation input is not traceable
L-16	Calculated ECM has no affected equipment
L-17	Simple payback lacks valid cost and savings
L-18	Avoided hours documented
L-19	ECM-02 — Economizer Repair: savings are not quantified.